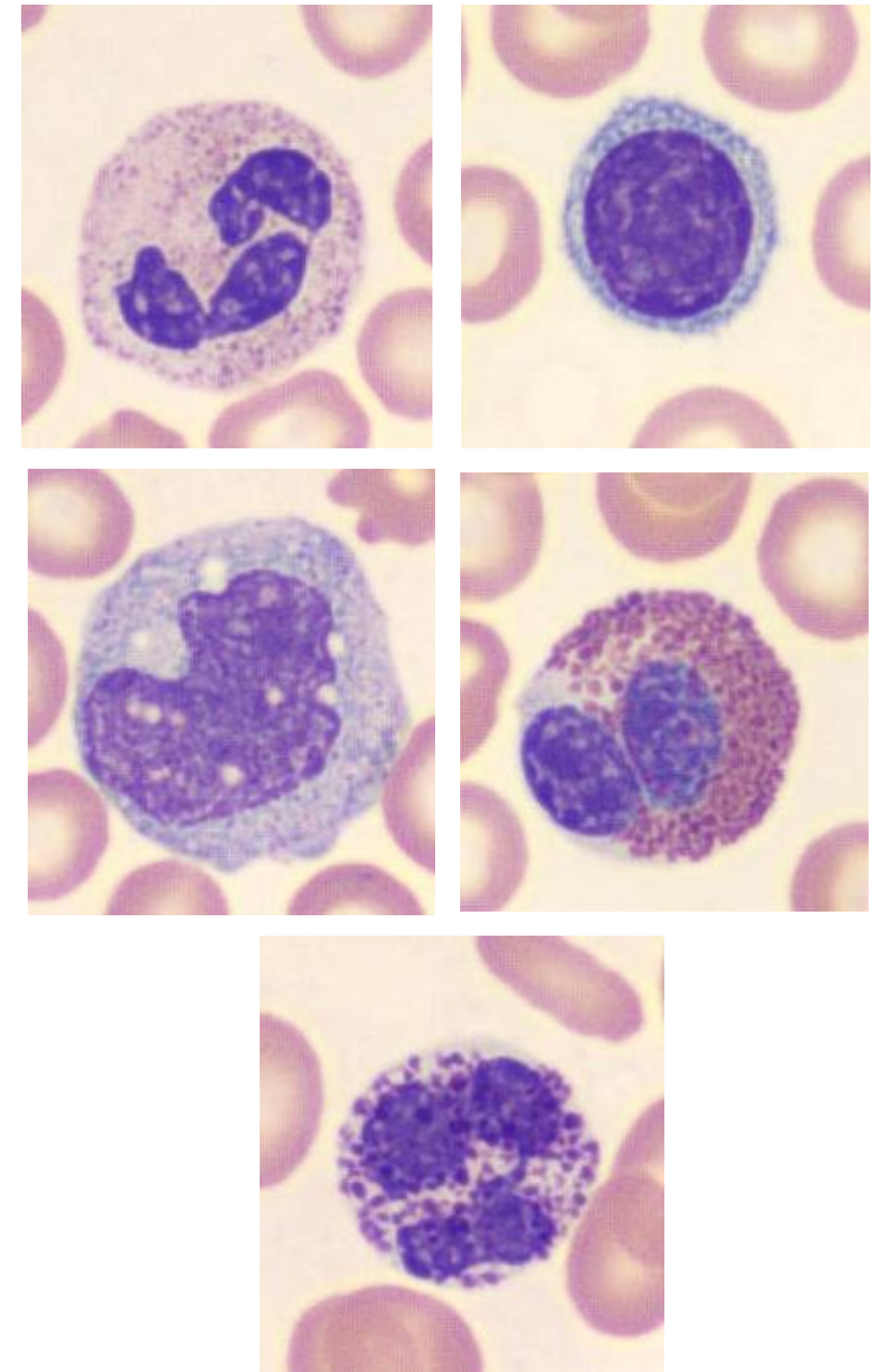


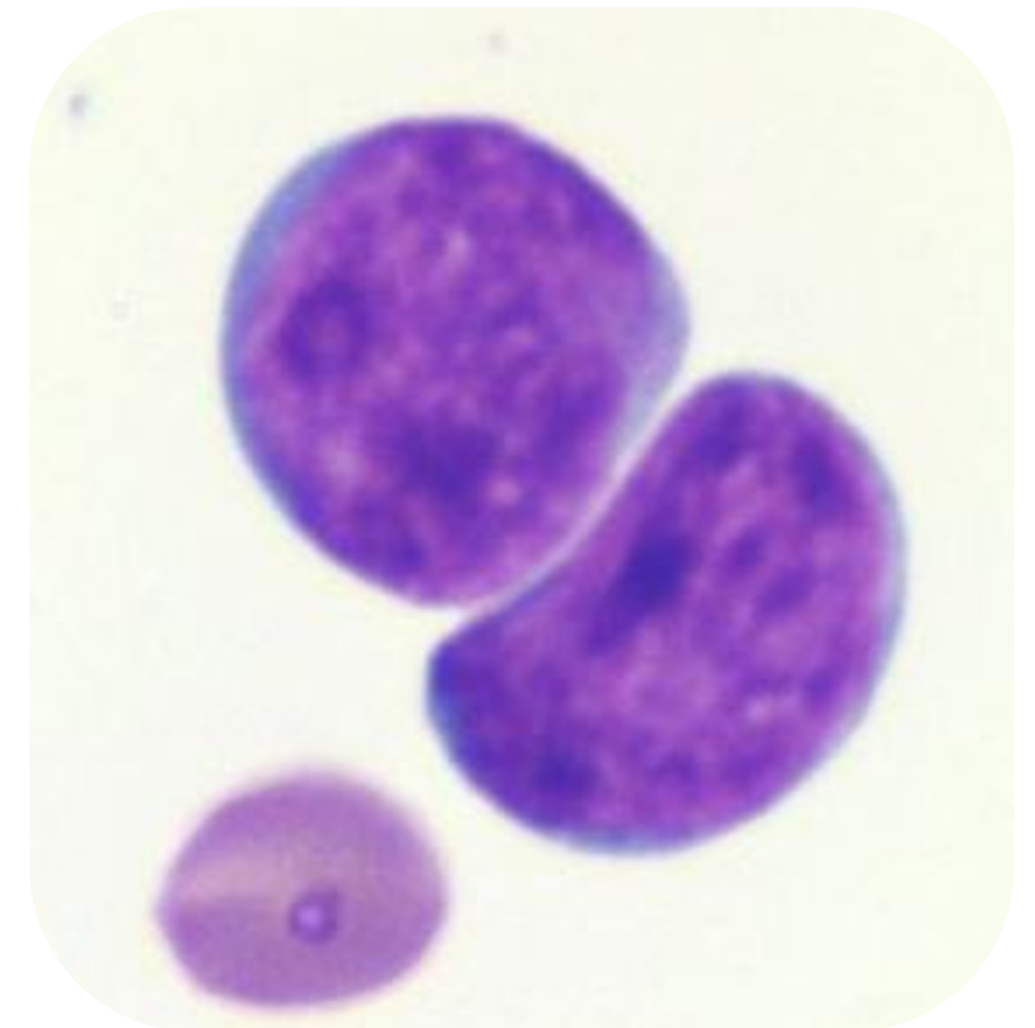
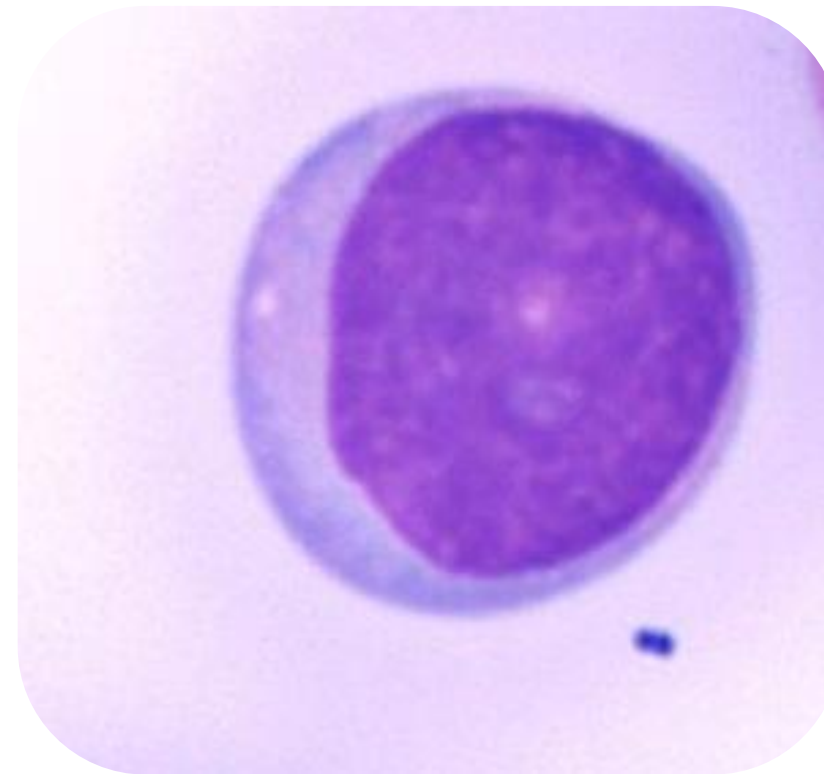
Lymphocytes

- ~20-40% of total WBC
- Subsets: B, T, NK
- Maturation:
 1. [Lympho]blast
 2. Prolymphocyte
 3. Lymphocyte



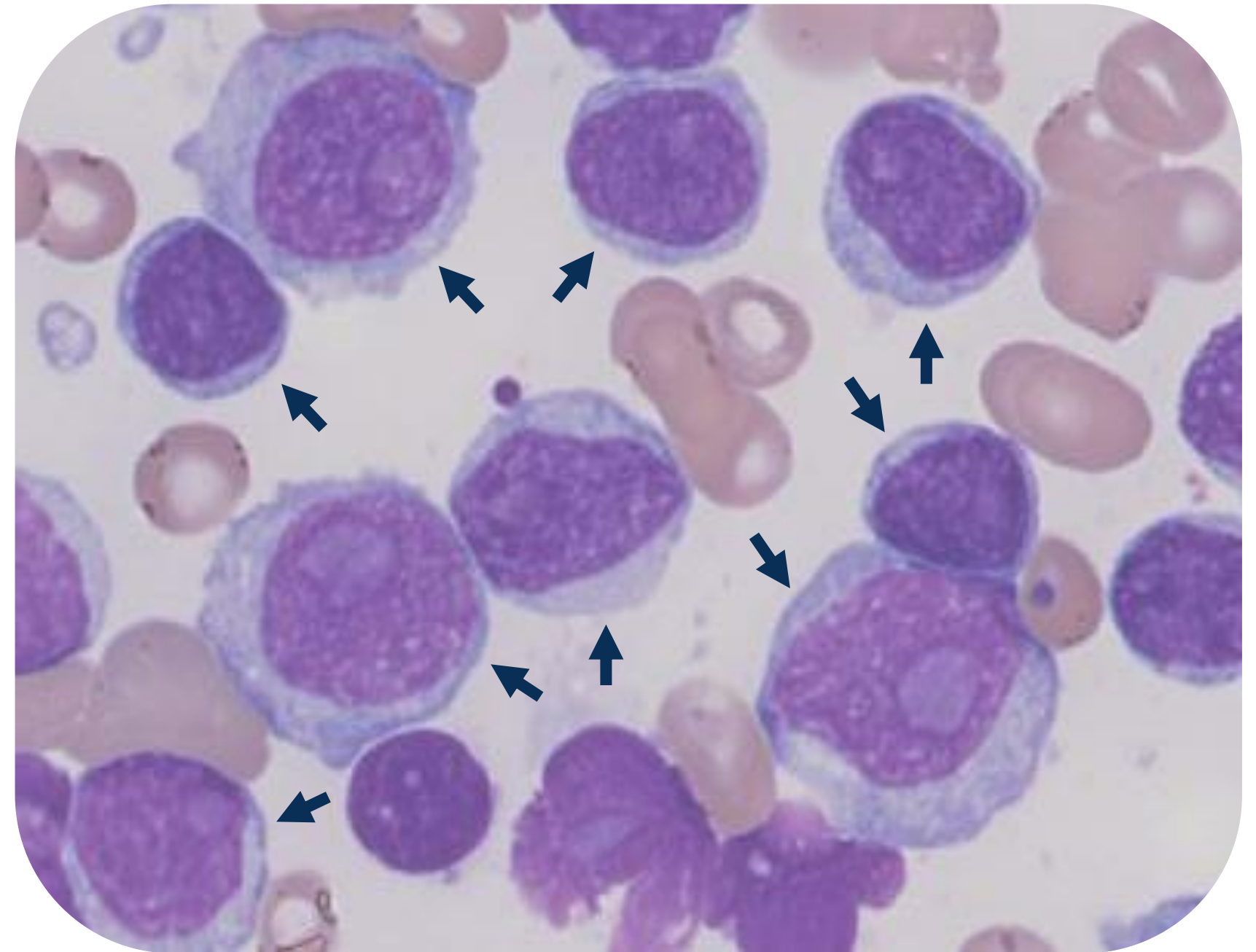
Blast (Lymphoblast)

- Nucleus
 - 10-18 μm , $\uparrow\uparrow\uparrow$ N:C
 - Lacy, fine, smudged, 1-2 nucleoli
- Cytoplasm
 - Agranular
 - [PAS, TdT] +
- Surface CD markers:
 - CD 34+ and CD 45+



Prolymphocyte

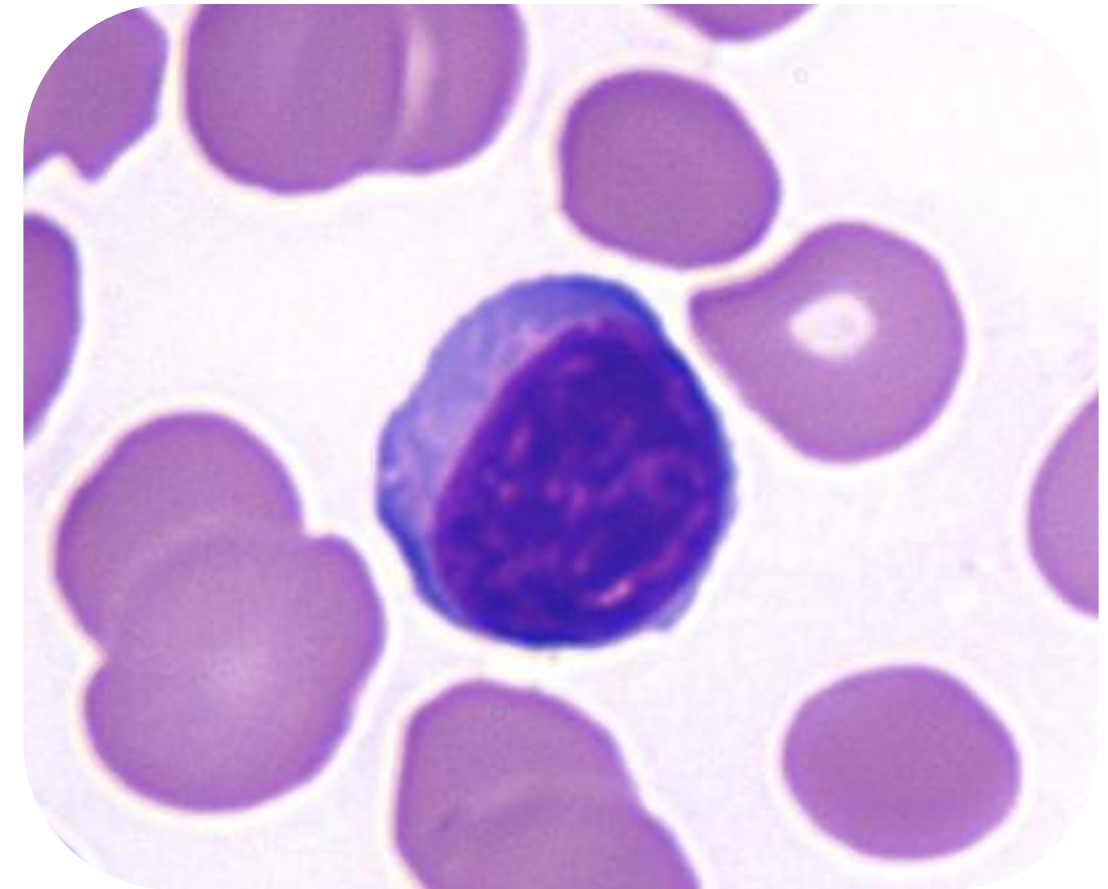
- Nucleus
 - Slight smaller than lymphoblast
 - ↑↑ N:C
 - Chromatin clumped/fine, nucleoli
- Cytoplasm
 - Light blue
 - Agranular



Note: This image is from a Prolymphocytic Leukemia so these prolymphocytes are larger than a normal; otherwise the morphology is similar

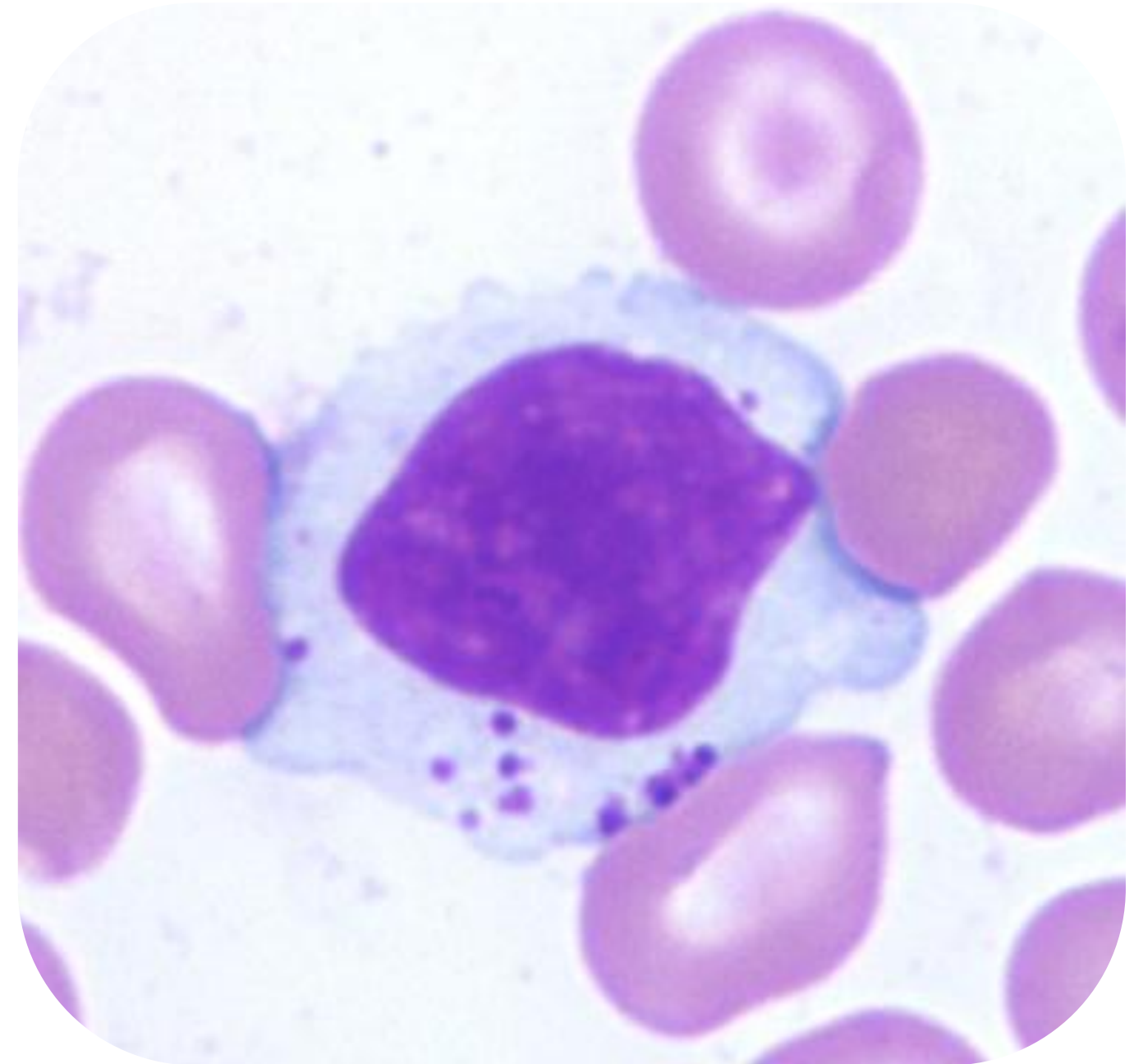
Lymphocyte

- Nucleus
 - Size of RBC, ~90% of cell
 - ↑↑ N:C
 - Chromatin condensed, nucleoli rare
- Cytoplasm
 - Sky blue
 - Granules possible



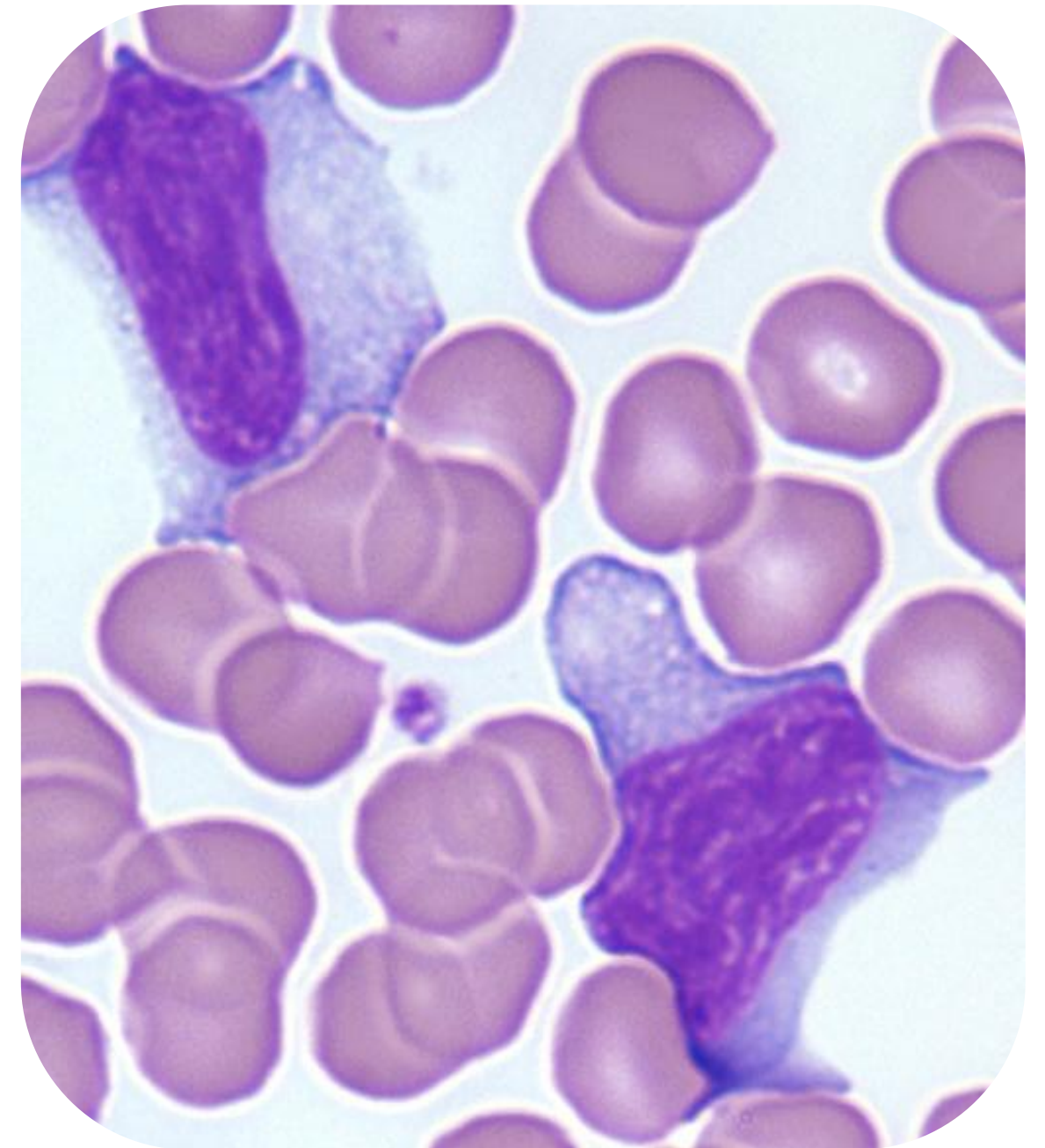
Large [Granular] Lymphocyte

- Mixed T/NK cells
- Nucleus
 - Round, indented
 - ↑↑ N:C
 - Chromatin condensed, nucleoli rare
- Cytoplasm
 - Pale blue
 - [Large, coarse granules possible]

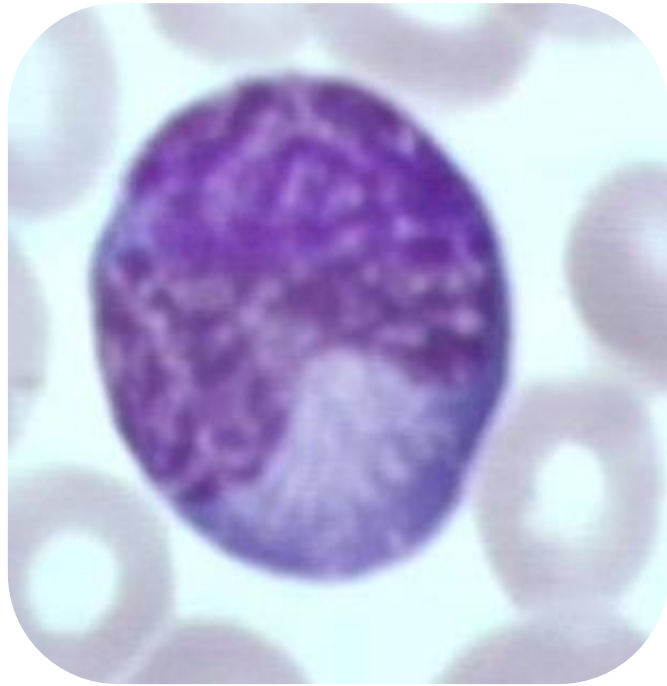


Reactive Lymphocytes

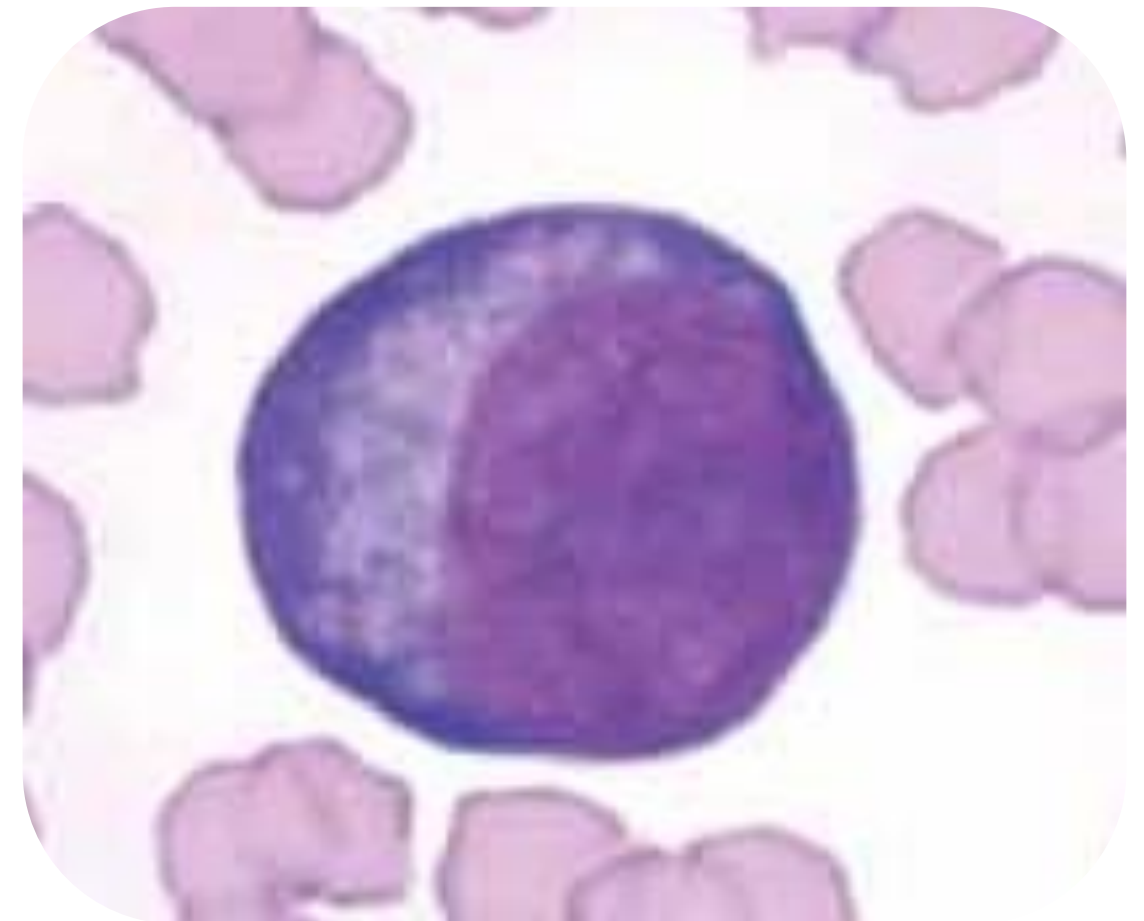
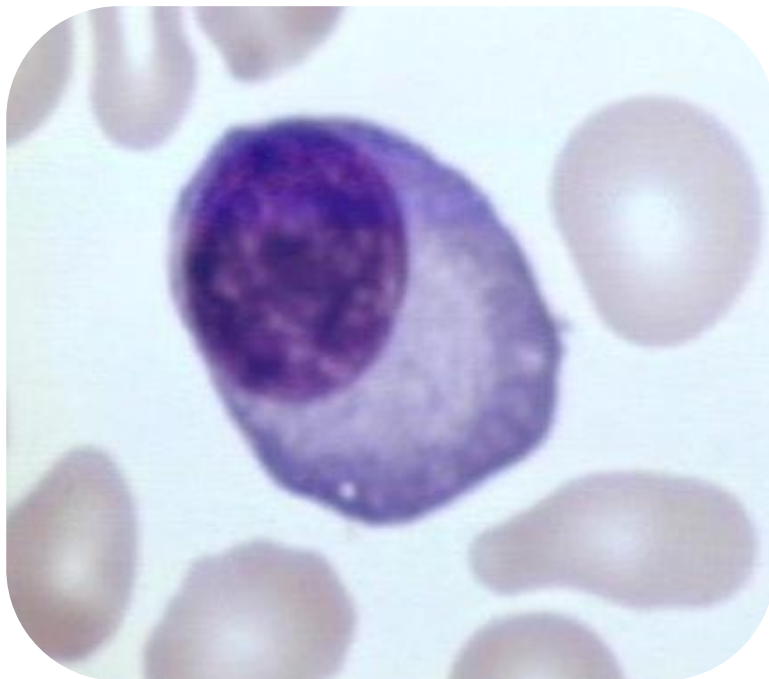
- B or T cells
- Nucleus
 - 16-30 μm , $\downarrow$ N:C
 - Variable shape/chromatin texture
- Cytoplasm
 - Diffuse or localized basophilia
 - Granules/vacuoles possible
 - “Hug” RBCs



Plasmacytoid Lymphs/Plasma cells

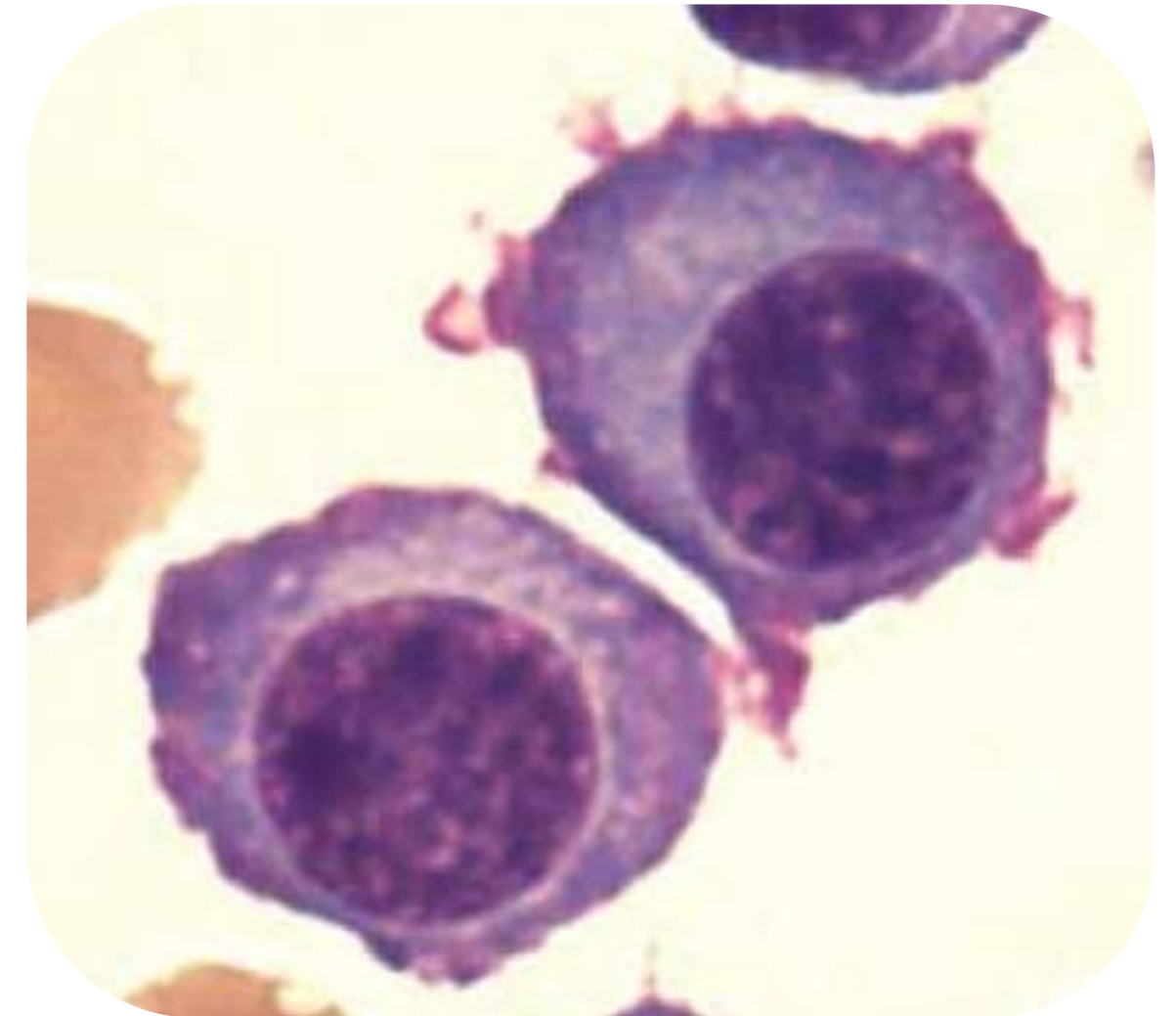
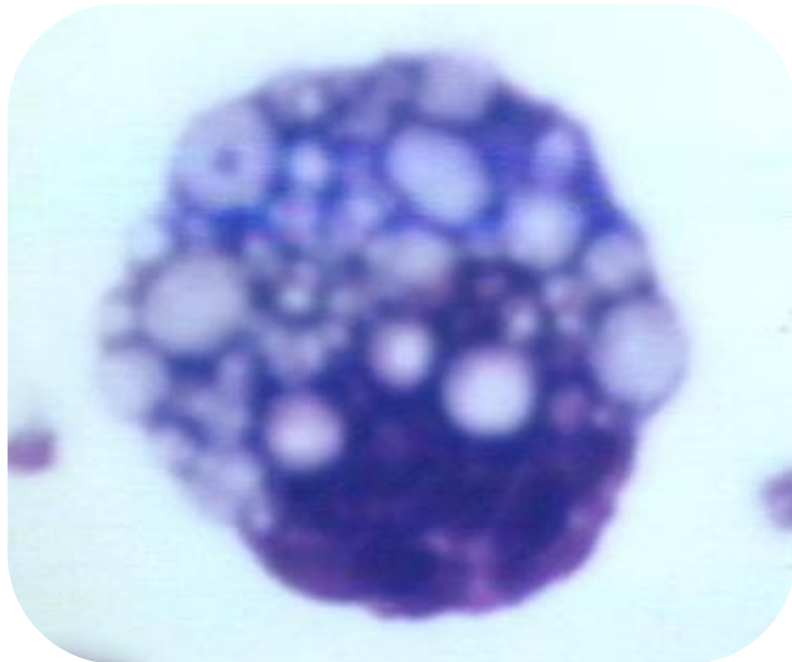


- Nucleus
 - 16-20 μm
 - Clumped chromatin
 - Eccentric
- Cytoplasm
 - Deep basophilia
 - Golgi apparatus



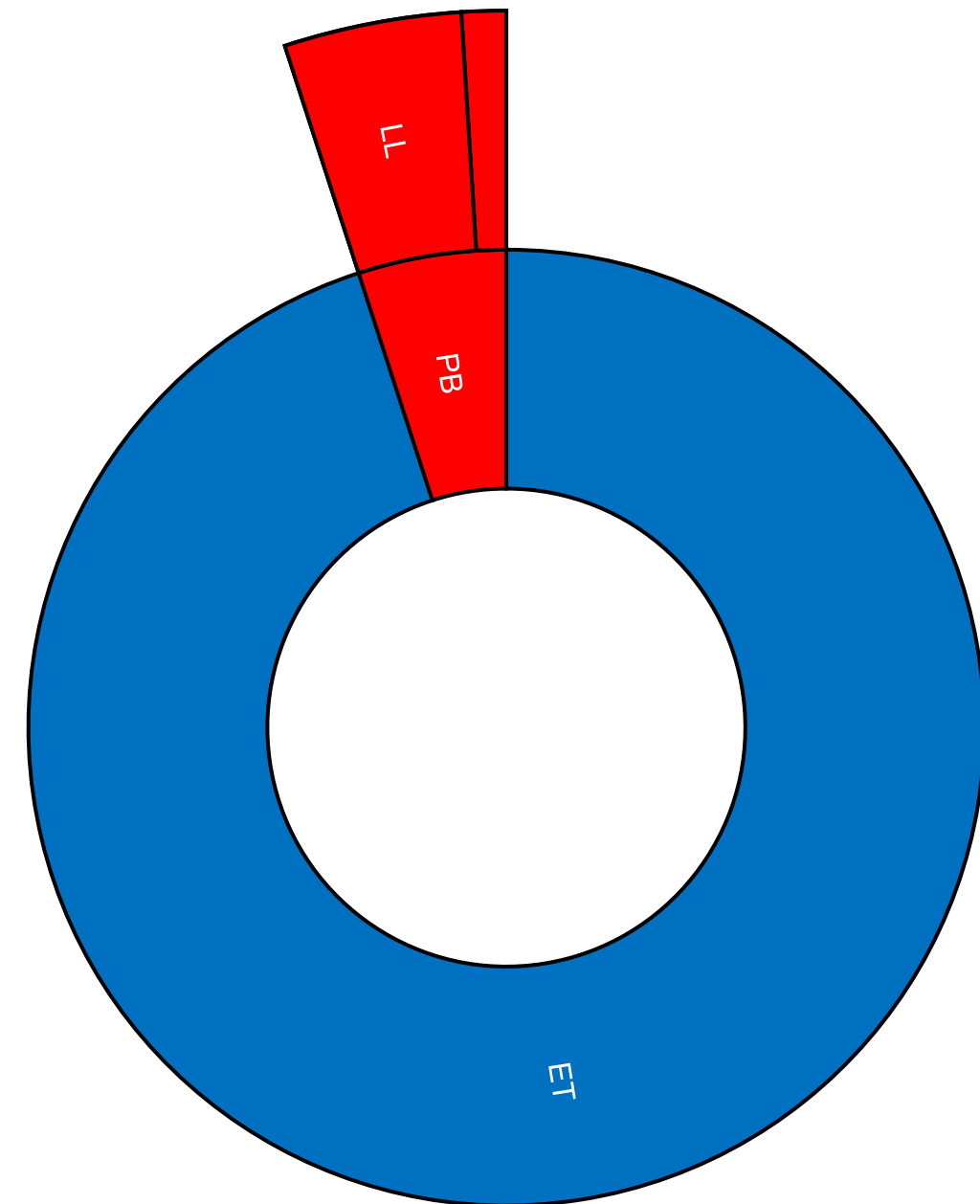
Plasma Cell Variations

- Flame Cell
 - Reddish-purple cytoplasm
 - More Ig
- Mott Cell
 - Russell bodies
 - Ig filled globules



Lymphocyte Kinetics

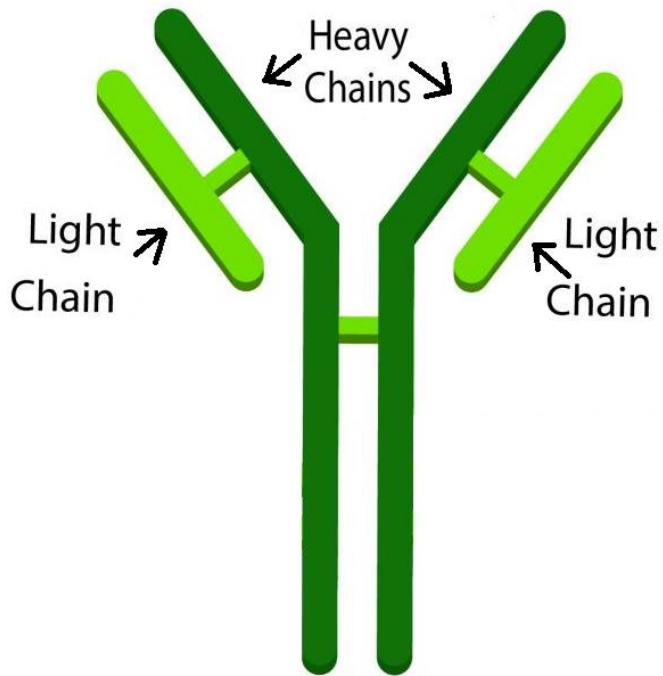
- Children greater concentration than adults
 - ~5% in PB, ~95% in extravascular tissue
 - ~80% in PB “long-lived”
 - ~20% hours to 5 days
- Lymphocytosis [$> 4.8 \times 10^3/\mu\text{L}$]
- Lymphocytopenia [$< 1.0 \times 10^3/\mu\text{L}$]



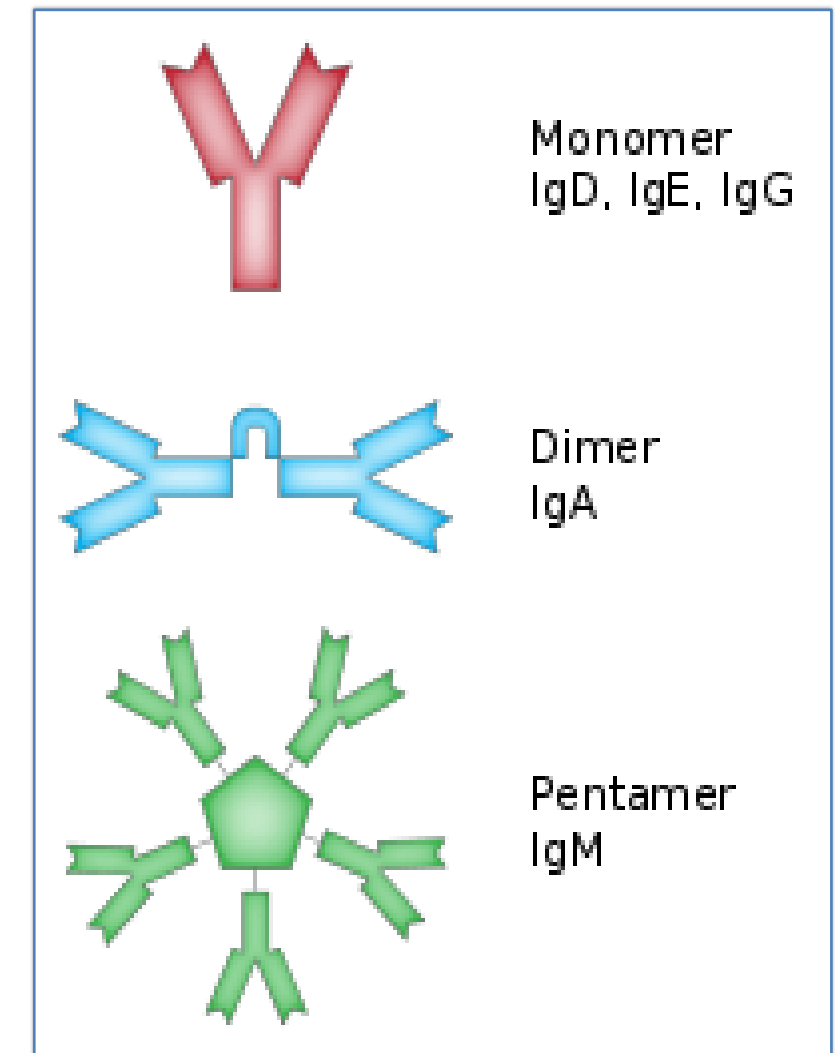
Lymphocyte Function

- Principal effector cells in immune viral response
 - Humoral (antibody-mediated) immunity
 - B cells become immunocompetent
 - Cell-mediated immunity
 - T_H (CD4+/8-), T_C (CD4-/8+), T_{Reg} cells
 - Effector cells of innate immunity
 - NK (CD16+/56+)
 - Activate macrophages

B Lymphocytes



- ~15-30% of PB lymphocytes
- [CD10, CD19, CD20]⁺
- Immunoglobulin (Ig)
 - Two identical heavy chains (IgA, IgD, IgE, IgG, IgM)
 - Two identical light chains (kappa, lambda)
- Primary vs secondary response
 - IgM (naïve B cells) vs IgG (memory B cells)
- Plasma cell is activated/terminal form



T/NK Lymphocytes

- Interact with APCs – macrophages and other T cells
- Secrete cytokines to activate B cells and macrophages
- [CD3, CD4]+ T cells (T helper)
 - MHC class II – extracellular pathogens
- [CD3, CD8]+ T cells (T cytotoxic/suppressor)
 - MHC class I – intracellular pathogens
- [CD16, CD56]+ NK for innate immunity
 - Kill infected cells, activates macrophages to destroy